COP 3515 – Advanced Program Design

Semester Programming Project

Client Change Request (CCR-007)

Project: Student Information Management System (SIMS)

Customer: University Registrar's Office

Priority: Medium

Assigned To: Student Developer

Sprint: Week 7

Due Date: Beginning o Lesson 16

Business Background

The University Registrar's Office has continued expanding the Student In ormation Management System (SIMS). Employees can now enter student in ormation, record grades, determine academic standing, save student records, and navigate the application through a menu-driven inter ace.

As more students are processed throughout the day, department administrators have requested summary in ormation that describes the entire day's activity rather than only the currently displayed student. Currently, employees have no easy way to determine how many students have been entered, the average GPA o all students processed, or how many students have earned Honors status.

The Registrar would like the application to maintain several statistics while the program is running. Some o this in ormation should be available throughout the entire application, while other in ormation should exist only while individual operations are being per ormed.

This enhancement should be added to the existing SIMS application developed during **CCR-001 through CCR-006**. **Do not create a new program.** Continue enhancing your existing codebase.

Business Requirements

The Registrar would like the application to continue performing **all functionality from CCR-001 through CCR-006**.

In addition, the application shall maintain the following statistics while the program is executing:

- Total number of students entered during this execution
- Average GPA of all students entered
- Total number of Honors students

The application shall include a new menu option:

6. Display Statistics

Selecting this option shall display the current program statistics.

The Registrar also wants the software to be designed so that information with different lifetimes is stored appropriately. Information that must remain available throughout the program should remain available, while temporary information should only exist while individual operations are executing.

Acceptance Criteria

The project will be considered complete when all of the following requirements have been met.

☐ The program compiles without errors or warnings.

☐ All functionality from **CCR-001 through CCR-006** continues to work correctly.

☐ The application correctly tracks the total number of students entered.

☐ The application correctly calculates the average GPA for all students entered.

☐ The application correctly tracks the number of Honors students.

☐ The new **Display Statistics** menu option functions correctly.

☐ Appropriate use of **local variables** and **global variables** is demonstrated.

☐ Output is neatly formatted and professional.

Technical Constraints

For this week's assignment, you may **only** use C language features covered through **Lesson 16**.

Permitted concepts include:

- Variables
- Constants
- Arrays
- Enumerations
- Loops
- Functions
- File processing
- Local variables
- Global variables
- Variable scope
- User input
- Formatted output

Do **not** use pointers or language features that have not yet been introduced in class.

Out of Scope

The Registrar is **not** requesting any of the following features at this time.

- Multiple student records stored simultaneously
- Searching student records
- Editing student information
- Deleting student records
- Dynamic memory allocation
- Databases

- Linked lists
- Graphical user interfaces (GUI)

These features will be requested in future Client Change Requests.

Questions for the Customer

Before writing any code, identify questions you would ask the customer to clarify the requirements.

Examples include:

1. Should the statistics be reset each time the program starts?
2. Should statistics include students loaded from a file or only students entered manually?
3. Should editing a student's GPA update the statistics?
4. How should the average GPA be displayed (number of decimal places)?
5. If the same student is entered twice, should that count as one student or two?
6. Should Honors students be counted when they are first entered or whenever their GPA changes?
7. Should statistics be saved to a file when the program exits?
8. Should the statistics be displayed automatically after entering a student?
9. Should the statistics include invalid student entries?
10. Which variables should remain global, and which should remain local?
11. Will future versions require additional statistics?
12. Should the statistics be available from every menu option?

Example Program Execution

Updated Main Menu

```
----------------------------------------
Student In□ormation Management System
Version 7.0
----------------------------------------

1. Add Student
2. Display Student
3. Enter Grades
4. Save Student Record
5. Exit
6. Display Statistics

Selection:
```

Example 1 – Enter Students

User Input

Selection: 1

Student ID: 1001

Student Name: Alice Johnson

Current GPA: 3.80

Selection: 1

Student ID: 1002

Student Name: Michael Brown

Current GPA: 2.95

Selection: 1

Student ID: 1003

Student Name: Sarah Smith

Current GPA: 3.65

Example 2 – Display Statistics

User Input

Selection: 6

Expected Output

SIMS Statistics

Total Students Entered : 3

Average GPA : 3.47

Honor Students : 2

Example 3 – Program Startup

Immediately a□ter the application starts:

```
----------------------------------------

SIMS Statistics

----------------------------------------


Total Students Entered : 0


Average GPA          : 0.00


Honor Students       : 0
```

Software Engineering Requirement

This Client Change Request introduces the concept o **variable scope**.

As the so tware becomes larger, it is important to care ully decide where variables should be declared.

Your solution should demonstrate an understanding o

- Variables that should remain available throughout the entire program (**global variables**).
- Variables that should exist only while a unction is executing (**local variables**).

Avoid declaring variables globally unless they truly represent in ormation that is shared across the application.

Regression Testing Requirements

Because the Student In ormation Management System is developed incrementally, **all previously implemented functionality must continue to operate correctly**.

Your submission must demonstrate that:

- CCR-001 through CCR-006 unctionality still works.
- CCR-007 correctly maintains application statistics.

At a minimum, include the ollowing regression tests.

Test	Description	Expected Result
1	Add one student	Total students = 1
2	Add three students	Total students = 3
3	Add Honors student	Honors count increases
4	Add non-Honors student	Honors count unchanged
5	Veri y average GPA calculation	Correct average displayed
6	Display statistics	Correct totals displayed
7	Save student record	Previous unctionality still works

Test	Description	Expected Result
8	Display student in□ormation	Previous □unctionality still works
9	Exit program	Program terminates normally

For **each** regression test include:

- Test description
- Input
- Expected output
- Actual output
- Pass/Fail result

I□a de□ect is discovered:

1. Document the de□ect.
2. Correct the de□ect.
3. Repeat the affected test.
4. Veri□y that all previously implemented □unctionality continues to operate correctly.

AI Usage Policy

Artificial Intelligence tools (such as ChatGPT, GitHub Copilot, Claude, or Gemini) may be used during this assignment.

However, you remain responsible for:

- Understanding every line of submitted code.
- Correctly selecting between local and global variables.
- Testing all functionality.
- Correcting any defects.
- Demonstrating that new functionality does not break existing features.

Your grade is based on the correctness, organization, and reliability of your software.

Submission Requirements

Submit a single ZIP file containing:

- main.c
- Any additional source (.c) files
- Any header (.h) files
- student_records.txt (generated by your program)

Also include a PDF containing:

- Screenshots demonstrating each menu option
- Screenshots of the **Display Statistics** feature
- Results of all required regression tests
- A brief description of any bugs encountered and how they were corrected
- Answers to the **Questions for the Customer**
- A brief explanation describing why each global variable was declared globally instead of locally